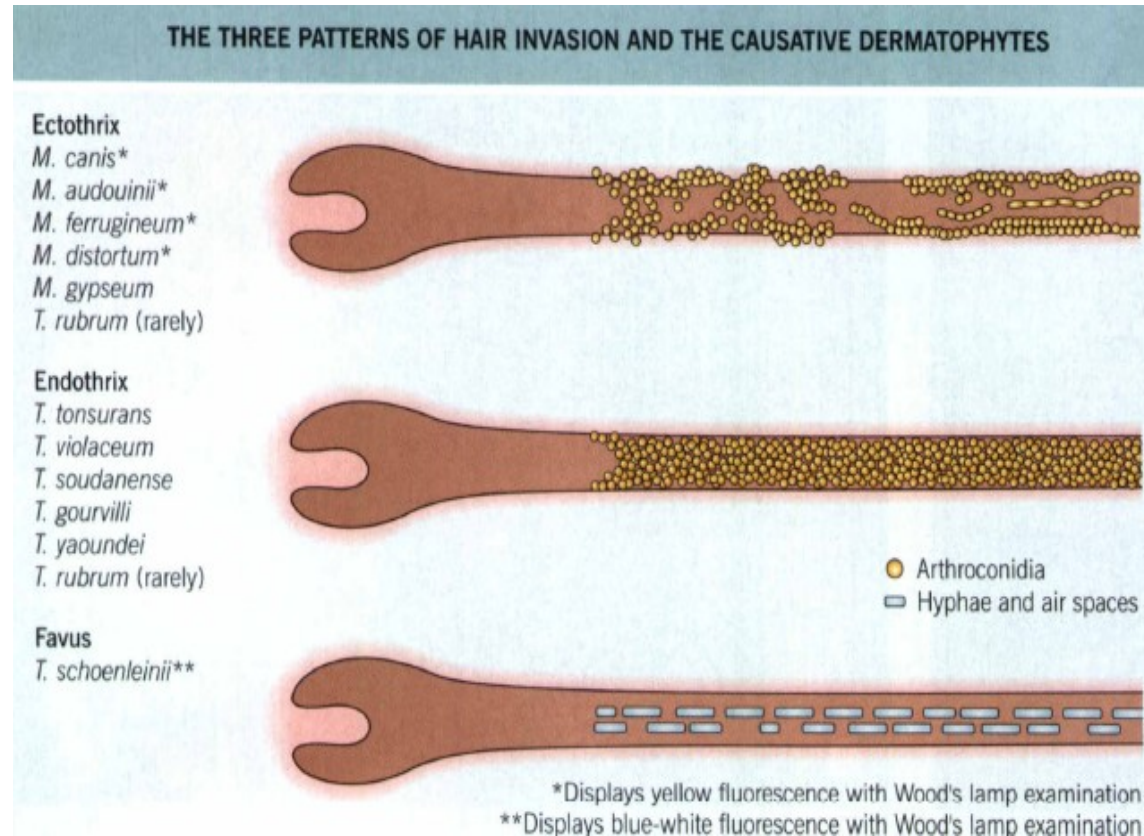




# Tinea Capitis

- Is a dermatophytosis of the scalp and associated hair.
- It may be caused by any pathogenic dermatophyte from the genera *Trichophyton* and *Microsporum* except *T. concentricum*
- The most common cause worldwide is *M. canis*
- **Ectothrix dermatophytes** typically establish infection in the perifoilicular stratum corneum
- **Endothrix infections** is the same except that the pathogen remain within the hair shaft, replacing the intrapilary keratin and leaving the cortex intact.
- Affects toddlers and school age children, sometimes as epidemics. Rare after 16 years.
- Transmission occurs from person to person, animal to person, or via fomites
- Poverty, debilitating chronic illness and malnutrition are risk factors



- It could be inflammatory or non inflammatory based on species involved.
- **C/F-**

### 1. Grey Patch:

- patches of scalp alopecia with scale. Small patches coalesce, forming larger patches. Inflammatory response minimal but massive scaling.



## 2. Black Dot:

- Broken-off hairs near surface give appearance of “dots”. Tends to be diffuse and poorly circumscribed. Resembles seborrheic dermatitis



### 3. Kerion:

- Characterized by boggy, purulent, inflamed nodules and plaques extremely painful; drains pus from multiple openings, like honeycomb

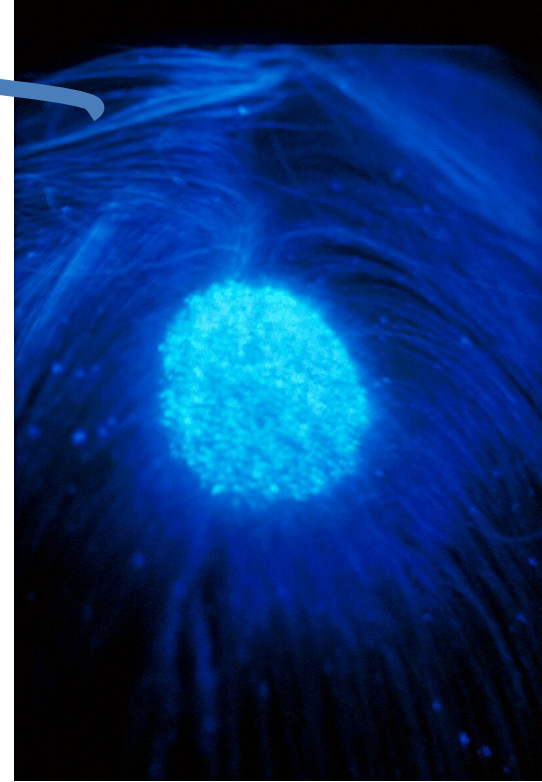


- **Favus:**
  - Thick yellow adherent crusts (scutula) composed of skin debris and hyphae that are pierced by remaining hair shafts. Fetid odor.



## Ixn-

- ✓ *Woods Lamp examination:*
  - green yellow fluorescence for lesions caused by *Microsporum spp*,
  - *T. Schoenleini* give dull green fluorescence
- ✓ *KOH microscopy*
- ✓ *Sabaroud's Agar Culture: up to 4 weeks at 26° C.*



## Mgt

- ✓ griseofulvin - *Pediatric Dose: 10-15(20-25) mg microsize/kg/d PO or 7.3 mg ultramicrosize/kg/d for 6-8 wk.* Better absorption with fatty food.
- ✓ Topical agents are ineffective.
- ✓ All contacts and family members should be examined and treated.
- ✓ Chronic untreated kerion and favus, can result in permanent scarring alopecia
- ✓ Steroids may help in inflammatory types (1mg/kg for 10-15 days)
- ✓ Selenium sulphide shampoo may help in decrease the risk of spreading infection.
- ✓ **Regrowth of hair is the rule if treated with systemic antifungal agents**

# Tinea Corporis

- Dermatophyte infections of the trunk, legs, and arms, excluding the feet, hands, and groin.
- Occurs in all ages as single (occasionally multiple) circular arcuate lesions with active raised margin and a tendency to clear centrally.
- Mostly asymptomatic
- Granulomatous reactions (**Majocchi's Granuloma**) and bullae formation with pruritis may occur.



## **Mgt**

- ✓ White field's ointment, topical azoles(1% Clotrimazole, 2%Ketoconazole) treats localized cases effectively.
- ✓ Should be given continuously till the lesion clears completely, at least three weeks.
- ✓ Systemic antifungals for extensive cases.

# Tinea Facialis

- ✓ seen on the **nonbearded parts of the face**
- ✓ The lesions are usually **pruritic**, itching and burning may become worse after exposure to sunlight.
- ✓ Some lesions may resemble those of tinea corporis; others may have little or no scaling or raised edges.
- ✓ Well-circumscribed erythematous patch
- ✓ Direct infection or secondary spread from preexisting infection site
- ✓ Commonly seen in children

## Etiologies

- ✓ T. mentagrophytes var. mentagrophyte and T. rubrum
- ✓ M. canis and M. audouinii

## Diagnosis & Treatment

- ✓ *As tinea corporis*







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# Oral Thrush

## *Classification of Candidiasis*

- **SUPERFICIAL MUCOSAL CANDIDIASIS**
  - Vaginal and Vulvovaginal Candidiasis
  - Oropharyngeal Candidiasis
    - Erythematous (atrophic) candidiasis
    - Pseudomembranous candidiasis (thrush)
    - Candidal leukoplakia (hyperplastic candidiasis)
    - Angular cheilosis
- **DEEP MUCOSAL CANDIDIASIS**





## **Mgt**

- ✓ Systemic or topical azoles.
- ✓ Nystatin BID is very effective for oropharyngeal candidiasis
- ✓ Chronic suppressive doses are needed for recurrent cases

# Atopic Dermatitis

- ✓ **Primary Defect** is synthesis of IgE rather than IgG, it is a **T cell - mediated hypersensitivity**
- ✓ The most common chronic relapsing skin disease in infancy and childhood
- ✓ Affects 10-30% of children
- ✓ **Atopic March**, sometime called

**Allergic March**, refers to the natural history or typical progression of **allergic** diseases that often begin early in life. These include **atopic** dermatitis (eczema), food allergy, **allergic** rhinitis (hay fever) and asthma.





## Clinical Manifestations

- begins in infancy, ~50% in the 1<sup>st</sup> year additional 30% are dx b/w 1-5 yrs.

## Cardinal Features

- \* Intense pruritis, specially at night & autaneous reactivity

↓  
scratching  
excoriation

$$\begin{array}{r} 280 \\ 2 \\ \hline 140 \end{array}$$

## Exacerbants / Triggers

- FOOD (cow milk, egg, peanut, tree nuts, soy, fish, wheat)
- Aeroallergens (Pollen, grass, animal dander, dust mites)
- Infections (Staphylococcus, HSV, molluscum)
- ↓ humidity, excessive sweating
- Irritants (wool, soaps, toiletries, fragrances...)

- \* Lesions: intensely pruritic, erythematous papules (Acute)

Erythematous, excoriated, scaling papules (Subacute)

Lichenification, skin thickening, fibrotic papules (Chronic)

- All the 3 types of skin reaction can be there at the same time in a pt

- \* Most pts have dry skin

## Distribution

Infants: symmetrically over the cheeks,  
forehead, scalp, trunk &  
extensor surface

- \* sometimes it  
can extensively  
involve flexural area

→ in severe forms  
it can cause  
alopecia

**Spares diaper area**

Children: symmetrically over the wrists, ankles,  
flexural areas

⇒ Generalized eruption can occur

## Diagnosis

AD is dx on the basis of 3 major features

- 1) Pruritis
- 2) Eczematous dermatitis  
that fits into typical  
presentations
- 3) Chronic & relapsing

Associated features = Family hx of asthma, allergic rhinitis, ↑ IgE  
hay fever

TABLE 3-1 -- Eczematous Inflammation

| Stage    | Primary and secondary lesions                                                      | Symptoms                                         | Etiology and clinical presentation                                                                                                                                   | Treatment                                                                                                            |
|----------|------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Acute    | Vesicles, blisters, intense redness                                                | Intense itch                                     | Contact allergy (poison ivy), severe irritation, id reaction, acute nummular eczema, stasis dermatitis, pompholyx (dyshidrosis), fungal infections                   | Cold wet compresses, oral or intramuscular steroids, topical steroids, antihistamines, antibiotics                   |
| Subacute | Redness, scaling, fissuring, parched appearance, scalded appearance                | Slight to moderate itch, pain, stinging, burning | Contact allergy, irritation, atopic dermatitis, stasis dermatitis, nummular eczema, asteatotic eczema, fingertip eczema, fungal infections                           | Topical steroids with or without occlusion, lubrication, antihistamines, antibiotics, tar                            |
| Chronic  | Thickened skin, skin lines accentuated (lichenified skin), excoriations, fissuring | Moderate to intense itch                         | Atopic dermatitis, habitual scratching, lichen simplex chronicus, chapped fissured feet, nummular eczema, asteatotic eczema, fingertip eczema, hyperkeratotic eczema | Topical steroids (with occlusion for best results), intralesional steroids, antihistamines, antibiotics, lubrication |

## Ddx

- seborrheic dermatitis
- contact dermatitis
- Nummular eczema
- Scabies
- Ichthyoses — fish skin
- HIV-associated dermatitis

## Management

- skin hydration = Moisturizers are 1<sup>st</sup> line therapy
- topical anti-inflammatory therapy
- Identify & eliminate flare factors
- Systemic Rx if necessary

① 15-20 min lukewarm soaking bath followed by occlusive emollient  
pts w AD have impaired skin barrier function from low lipid levels  
so they have xerosis = dry skin

② Topical corticosteroids are the cornerstone of anti-inflammatory Rx  
for acute exacerbations of AD.

\* There are 7 groups of drugs according to their potency

+ Natural sunlight is often beneficial = PHOTOTHERAPY

Ddx: Seborrhoeic eczema

- Infantile
- Adult

Complications

Infection:

- Bacterial:
  - Staphylococci
- Viral:
  - Herpes viruses
  - molluscum contagiosum
  - Fungi – pityrosporum orbiculare

# Seborrheic Dermatitis(SD)

- ✓ A very common chronic, relapsing, and usually mild form of dermatitis that occurs in infants and in adults.
- ✓ The severity may vary from minimal, asymptomatic scaliness of the scalp (dandruff) to more widespread involvement
- ✓ SD has a biphasic incidence, occurring in infants between the ages of 2 weeks and 12 months, and later, during adolescence and adulthood.

## **Predisposing and Exacerbating Factors**

- Genetic
- Immunosuppression - HIV infection
- Parkinson disease
- Emotional stress

**Etiology :** Fungus of genus *Malassezia*

**C/F-** usually characterized by well demarcated erythematous plaques with greasy-looking, yellowish scales distributed on areas rich in sebaceous glands.



- The mildest and most common form of scalp seborrheic dermatitis is dandruff
- Dandruff may be asymptomatic or accompanied by mild pruritus



- Periocular — Blepharitis with redness of the free margin of the eyelids and yellow crusting between the eyelashes may be the sole manifestation of seborrheic dermatitis.
- Face — Facial lesions favor the forehead below the hairline, the eyebrows and glabella, and the nasolabial folds. They may extend to the cheeks and malar areas in a butterfly distribution.



Dx - usually made clinically based on the appearance and location of the lesions.

Ddx - Psoriasis –in adolescents

Atopic dermatitis, SLE Malar rash, Rosacea, Tinea corporis/fasialis

Rx - **Initial Topical Therapy**

- **Scalp-** Effective over-the-counter (OTC) *shampoos* containing selenium sulfide, zinc pyrithione, are helpful.  
Tar shampoos (OTC) are equally effective in many patients. Low-potency *glucocorticoid* solution, lotion, or gels following a medicated shampoo (ketoconazole or tar) for more severe cases.
- **Face and Trunk-**  
*Ketoconazole shampoo* ,  
2% Glucocorticoid cream and lotions:  
Initially 1 or 2.5% hydrocortisone cream, 2% ketoconazole cream,  
1% pimecrolimus cream,  
0.03 or 0.1% tacrolimus ointment.
- **Eyelids** Gentle removal of the crusts in the morning with a cotton ball dipped in diluted baby shampoo. Apply 10% sodium sulfacetamide in a suspension containing 0.2% prednisolone and 0.12% phenylephrine.
- **Systemic Therapy**  
In severe cases, 13- *cis* retinoic acid orally, 1 mg/kg, is highly effective.  
In milder cases, itraconazole 100 twice daily for 2 weeks is also effective.

# Diaper Dermatitis

- **Diaper dermatitis** is a general term used to describe any inflammatory skin eruption that develops in the diaper-covered region.
- caused by overhydration of the **skin**, maceration, prolonged contact with urine and feces, retained diaper soaps, and topical preparations and is a prototypical example of irritant **contact dermatitis**. Signs and symptoms are restricted to the area covered by diapers.
- major categories of diaper dermatitis:
  - ✓ Rashes associated with or worsened by the diaper
  - ✓ Rashes that occur regardless of diaper use

## C/F -

- ✓ erythema
- ✓ painful scaling papules and superficial erosions
- ✓ Irritant diaper dermatitis typically occurs on convex skin surfaces that are in direct contact with the diaper. These locations include the buttocks, lower abdomen, genitalia, and upper thighs. The skin folds (areas not in direct contact with the diaper) are classically spared

# Classification of diaper dermatitis

| Associated with diaper use   |
|------------------------------|
| Irritant diaper dermatitis   |
| Candidal dermatitis          |
| Jacquet's erosive dermatitis |
| Granuloma gluteale infantum  |
| Allergic contact dermatitis  |

| Regardless of diaper use |
|--------------------------|
| Seborrheic dermatitis    |
| Atopic dermatitis        |
| Psoriasis                |
| Impetigo                 |
| Scabies                  |
| Langerhans histiocytosis |
| Nutritional deficiency   |
| Immunodeficiency         |
| Child abuse              |



Aspects of the history that can help identify contributing factors and support or exclude nondiaper-associated dermatitis include:

- Associated symptoms (eg, diarrhea).
- Systemic symptoms (may suggest nondiaper-associated dermatitis, particularly LCH).
- Information about diapers and diapering: type of diaper, how often diapers are changed, method of laundering cloth diapers (if cloth diapers are used).
- Information about how the diaper area is cleansed (eg, soaps, cleansers, washcloths, wipes, etc).
- Exposure to contagious disease (eg, scabies, herpes simplex virus, Staphylococcus aureus, group A Streptococcus).
- Past history of dermatologic, allergic, or infectious illnesses.
- Family history (eg, psoriasis, atopy).
- Antibiotic use (predisposes to candidal dermatitis and diarrhea).

**Ixn-** usually are not necessary. help confirm the etiology in atypical or recalcitrant cases:

- Potassium hydroxide preparation and/or fungal culture of skin scrapings for Candida
- Mineral Oil slide preparation for scabies
- Tzanck smear of blisters or pustules; multinucleated giant cells indicate a viral infection such as herpes simplex
- Serum zinc level for acrodermatitis enteropathica
- Culture of skin lesions for *S. aureus* or group A streptococcus
- Skin biopsy in cases in which the rash is atypical or unresponsive to therapy (for LCH, granuloma gluteale infantum)

**Management** — **eliminate direct skin contact** with urine and feces (ie, by discontinuing or limiting the use of diapers)

- **Topical Barrier** (Petroleum Jelly, Zn oxide cream, Paraffin)

- **Powders**

- **Antifungal agents** — nystatin, cotrimazole, miconazole, ketoconazole

are effective topical therapies for candidal diaper dermatitis. Antifungal agents are indicated when:

- ✓ The dermatitis appears to be candidal (eg, beefy red plaques, satellite papules, superficial pustules; involvement of the skin folds)
- ✓ Is confirmed to be candida by potassium hydroxide preparation or fungal culture
- ✓ The dermatitis has been present for at least three days (which increases the likelihood of secondary superinfection with candida).

- Antibiotics** — If a secondary bacterial infection is present